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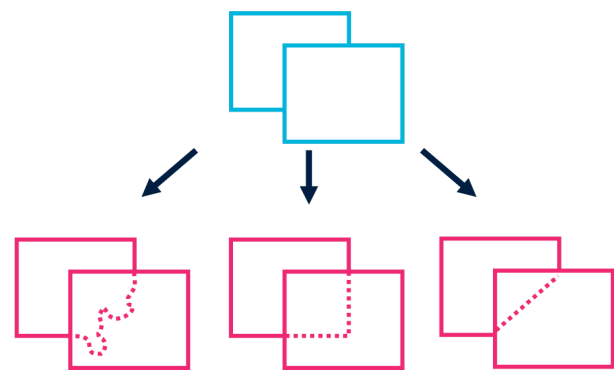
Does our brain measure Kolmogorov complexity?

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Question: What is the difference between Kolmogorov complexity and cognitive complexity?

Answer: Cognitive complexity is resource-bounded.

Several phenomena point to the fact that the human brain is sensitive to Kolmogorov complexity. [Nick Chater suggested](#) that simplicity should be used as a fundamental principle in Cognitive Science. Here is an example.



The preferred hidden pattern is not the complex one, but the closing curve that appears **simplest** (in the usual Kolmogorov sense). Note that the straight line, which would be the simplest closing curve in the absolute, is not the simplest one in the context. **Cognitive complexity** is a resource-bounded version of **description complexity**. It is defined as the shortest description (in bits) of the situation s that a human individual could reach at time t . In a semi-formal way:

Cognitive complexity

$$C_d(s) = \min_p \{ \text{length}(p); H^t(p) = s \}$$

Here, H represents the cognitive model that simulates the individual's behaviour. H^t represents the same model that is allowed to run during t units of time. p is any characterization that leads to the unambiguous determination of s and that is available to H^t . Note that $C_d(s)$ may vary through time. From now on, the name **description complexity** will be used to refer to its resource-bounded version: **cognitive complexity**.

Description complexity

1/1 point (ungraded)

If C represents plain complexity, what should we say?

☒ A. $C(s) \leq C_d(s)$.

☐ B. $C(s) \geq C_d(s)$.

☐ C. $C(s)$ and $C_d(s)$ cannot be compared.

✓

Answer

Correct:

We may consider that H in the definition of C_d is a universal Turing machine: given unlimited time and access to unlimited memory (such as notepads), human beings may perform any task that a computer can do. Since H^t is constrained, $C(s) \leq C_d(s)$.

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